

PersonaMetrica: Stress-Testing Evaluation Protocols for Long-Horizon Personal Agents

Anonymous submission draft — NeurIPS Evaluations & Datasets framing

Abstract

Long-horizon personal assistants must do more than retrieve a previously stated preference: they must determine which evidence still applies, represent uncertainty when evidence conflicts, respect temporary scope, and decide when a current belief is reliable enough to act on. Existing memory and personalization evaluations often emphasize recall, preference following, or state accuracy. We study a narrower evaluation question: **which conclusions are stable once confidence is allowed to control action, and which depend on the calibration protocol itself?** We introduce **PersonaMetrica-Bench**, a controlled evaluation framework that jointly measures current-state accuracy, confidence calibration, action coverage, selective accuracy, risk-coverage, and downstream decision utility under evolving preferences. We also introduce **Personal Belief State (PBS)**, a transparent evidence-weighted temporal reference system. The standard release contains **500 synthetic users, 27,133 observations, and 13,500 decision queries** across ten preference domains. On the standard controlled run PBS obtains **0.932 vs. 0.906 state accuracy, 0.060 vs. 0.080 Brier**, and lower AUC (0.0107 vs. 0.0208) than a strong **time-aware-latest** baseline. However, our most important result is negative: **fixed-threshold decision utility is not protocol-invariant**. With native confidence, PBS appears substantially better than time-aware-latest on 800-turn histories; after every tracker receives the same low-capacity post-hoc calibration on development seeds only, the long-horizon utility ranking reverses (**0.588 PBS vs. 0.687 time-aware-latest**) while PBS retains higher selective accuracy and slightly lower Brier. A separate frozen audit over 75 protocol cells and all six released baselines finds that time-aware-latest wins 45 cells, PBS 27, and first-mention 3; two protocol cells choose different winners with probability **0.516**, mean complete-ranking Kendall τ is **0.699**, and the most discordant pair falls to **0.20**. A paired seed-bootstrap diagnostic gives winner-change **0.516 [0.507, 0.538]**, and 13 leave-one-level-out grid perturbations all retain multiple winners. Thus neither raw state accuracy nor fixed-threshold utility from unmatched confidence scales is a safe standalone model-selection criterion. We recommend reporting state accuracy, calibration, risk-coverage/AUC, and utility only under an explicitly shared calibration-and-action protocol. PersonaMetrica-Bench releases six baselines, development/test splits, calibration and sensitivity analyses, component ablations, a human-annotation interface, an external-model evaluation contract, machine-readable metadata, and full reproduction artifacts. We explicitly do not claim frontier-model or human-population generalization from the controlled study.

1. Introduction

Personalization is commonly framed as a memory problem: a user states a preference, an assistant stores or retrieves it, and evaluation asks whether the assistant later recalls or follows it. That abstraction becomes incomplete once preferences evolve, evidence is noisy or implicit, temporary exceptions occur, and memory is used to trigger recommendations, reminders, tool actions, or other interventions.

Consider a user who generally prefers morning workouts, temporarily switches to evenings during a two-week schedule disruption, and later returns to mornings. A system that always uses the latest apparently trustworthy fact can be highly accurate on aggregate state queries yet become overly conservative or overconfident in exactly the cases where action requires calibrated uncertainty. A second system can maintain a distribution over plausible current values and decide whether to act or defer. These systems should not necessarily be judged by the same scalar recall metric.

This paper studies a stricter evaluation question: **when long-horizon personalization drives action, which conclusions survive changes in calibration protocol and evaluator assumptions?** We formulate long-horizon personalization as partially observed state estimation followed by risk-sensitive decision-making. PersonaMetrica-Bench separates the observation layer from the belief-update layer so the latter can be stress-tested with known provenance and scope. We then evaluate both correctness and the relationship between confidence and downstream action.

Our strongest result is not that one memory algorithm always wins. Rather, the ranking depends on **both the metric and the confidence protocol**. With native confidence, the 800-turn experiment makes PBS look like a

substantially better action substrate despite slightly lower state accuracy. Once every tracker is given the same development-only post-hoc calibration opportunity, the fixed-threshold utility ordering flips and time-aware-latest becomes better at that operating point. AURC, which depends on confidence ordering rather than a particular action threshold, still favors PBS. This exposes two evaluation confounds at once: state accuracy can miss selective-risk structure, while fixed-threshold utility can reward one system merely because its confidence scale is better matched to the chosen threshold.

Contributions

1. **Evaluation framing.** We formalize long-horizon personalization as belief-state estimation plus action under uncertainty and argue for reporting calibration and risk-coverage alongside state accuracy.
2. **PersonaMetrica-Bench.** A deterministic controlled testbed for protocol sensitivity with ten preference domains, durable changes, temporary exceptions, explicit/implicit evidence, noisy behavioral cues, hypothetical and other-person distractors, long-horizon variants, counterfactual interventions, and trajectory-level grader stress tests.
3. **Strong executable baselines.** Six memory/state systems ranging from first-mention and latest-observation to a time-aware baseline that filters distractors and handles temporary expiry.
4. **Personal Belief State.** A transparent evidence-weighted temporal representation with provenance, source reliability, recency, context scope, expiry, and probabilistic readout.
5. **Evaluation analysis.** Held-out multi-seed statistics, risk-coverage/AURC, a threshold/deferral-cost sensitivity sweep, and a development-only calibration protocol that reveals when native-confidence utility comparisons are not fair.
6. **Evaluator-as-system discipline.** Held-out grader attacks, causal trajectory contracts, abstention under missing evidence, implementation fingerprints, and disagreement routing test whether the measurement system itself generalizes. A 620-item human-annotation pack and vendor-neutral external-model evaluator are included as unexecuted infrastructure; no human or frontier-model result is fabricated.

Research questions

- **RQ1 — Metric sufficiency:** Does state accuracy rank personal-state systems the same way as selective risk and downstream action quality?
- **RQ2 — Calibration fairness:** Do conclusions from fixed-threshold utility survive when every system receives the same development-only confidence calibration opportunity?
- **RQ3 — Mechanism:** Which evidence-handling components of PBS account for its behavior under noise, temporary scope, and long histories?

The paper treats RQ2 as a first-class evaluation question rather than a post-hoc implementation detail. Any utility comparison that gates action on confidence is meaningful only relative to an explicit confidence calibration protocol.

2. Problem formulation

Let a user have latent preference state z_t . The agent observes evidence x_t , which may be explicit, implicit, noisy, hypothetical, about another person, or scoped to a temporary context. A personalization system maintains a belief

$$b_t(z) = p(z_t \mid x_{1:t})$$

and uses it to answer a query or choose an action a_t .

A conventional state metric measures

$$\text{Acc} = \mathbb{E}[\mathbf{1}(\hat{z}_t = z_t)].$$

For an acting assistant this is incomplete. We additionally measure:

- **coverage:** fraction of queries on which confidence is above an action threshold;
- **selective accuracy:** accuracy conditional on action;

- **Brier score:** calibration of confidence against correctness;
- **risk-coverage curve:** error rate as progressively less-certain decisions are included;
- **AURC:** area under the risk-coverage curve, lower is better;
- **decision utility:** a transparent operating-point metric in which correct action is +1, wrong action is -1, and deferral has an explicit cost.

The utility coefficient is not assumed to be universal. We therefore report a sensitivity sweep and treat AURC as the more defensible threshold-free measure of confidence quality.

3. Personal Belief State

PBS stores evidence rather than overwriting a single current value. Each evidence record includes:

- semantic slot and candidate value;
- turn and provenance;
- source type: explicit, implicit choice, implicit behavior, inferred, hypothetical, or other-person;
- observation confidence;
- optional context scope;
- optional temporary validity horizon.

At query time, each record receives a weight determined by source reliability, observation confidence, exponential recency, context compatibility, and temporary validity. Candidate values are normalized to a probability distribution. Hypothetical and other-person evidence remain in provenance but receive near-zero persistence weight. Temporary evidence is strong in scope and sharply discounted after expiry rather than erased.

PBS is intentionally transparent and dependency-light. It is not presented as a replacement for a learned LLM memory module. Its role is twofold: to instantiate the belief-state evaluation hypothesis and to provide an auditable reference system whose failure modes can be studied before introducing model-extraction errors.

3.1 Development-only parameter selection

PBS has two continuous parameters in the checked-in experiment: evidence half-life and softmax temperature. We select them by **minimum mean Brier score on development seeds 1–3 only**, using 120 users per seed. The selected values are a 90-turn half-life and temperature 0.20. All reported ten-seed standard statistics use held-out seeds 10–19, and long-horizon statistics use held-out seeds 30–39. The full search table is released in `reports/BELIEF_TUNING.md`.

4. PersonaMetrica-Bench

4.1 User state

The benchmark uses ten binary preference domains: work time, response detail, travel pace, restaurant noise, notification style, budget style, exercise time, meeting style, reading depth, and planning style.

4.2 Evidence processes

Each simulated user begins with explicit preferences and later receives 4–7 durable changes. Histories also contain:

- implicit-choice updates;
- repeated behavioral cues that are **noisy** rather than always correct;
- hypothetical distractors;
- statements about another person;
- temporary exceptions with explicit expiry;
- queries that may fall inside or outside temporary scope.

The standard exported dataset contains **500 users, 27,133 structured observations, and 13,500 decision queries**. Surface-language metadata spans English, Spanish, French, Persian, and Chinese. The current benchmark begins **after** evidence extraction, so these tags must not be interpreted as multilingual natural-language understanding results.

4.3 Distribution shifts

We evaluate five controlled regimes:

1. **Standard:** 500 users $\times$ 240 turns.
2. **High noise:** more hypothetical/other-person distractors.
3. **Implicit-heavy:** more preference changes conveyed via implicit evidence.
4. **Temporary-heavy:** substantially more scoped exceptions.
5. **Long horizon:** 180 users $\times$ 800 turns in the checked-in shift run.

5. Baselines

We compare six executable systems:

- **First-mention:** stores the first observed value forever.
- **Latest-observation:** blindly overwrites with the newest observation.
- **Sliding-window-40:** uses the newest observation in a 40-turn window.
- **Trusted-latest:** removes obvious hypothetical/other-person evidence but does not model temporary expiry or aggregate uncertainty.
- **Time-aware-latest:** a strong symbolic baseline that filters distractors, distinguishes general and temporary evidence, and checks expiry at read time, but does not aggregate competing evidence.
- **PBS:** evidence-weighted probabilistic belief state.

The inclusion of time-aware-latest is important: without it, the benchmark would reward a method merely for knowing explicit scope semantics. The repository records the earlier failure in which this baseline initially solved a cleaner version of the benchmark, motivating noisy behavioral evidence in the final generator.

These controlled baselines are not substitutes for model-level comparisons with LongMemEval, PrefEval, Horizon-Bench, PerMemBench, ProactiveBench, or production memory stacks. The release includes an external-model contract specifically so those results can be added without changing the metric implementation.

6. Metrics and statistical protocol

The primary controlled table reports state accuracy, action coverage, selective accuracy, decision utility at threshold 0.70 and deferral cost 0.10, Brier score, and general/temporary slices.

For threshold-free confidence quality, we report the risk–coverage curve and AURC. For sensitivity, thresholds $\{0.50, 0.60, 0.70, 0.80, 0.90\}$ are crossed with deferral costs $\{0, 0.05, 0.10, 0.20, 0.40, 0.60\}$.

For the standard PBS comparison, ten held-out seeds (10–19) use 220 users $\times$ 240 turns each. For the long-horizon ranking analysis, ten held-out seeds (30–39) use 90 users $\times$ 800 turns each. Bootstrap intervals summarize **seed-level simulator variation**, not uncertainty over a human population.

7. Results

7.1 Standard controlled benchmark

On the checked-in standard run:

Method	State acc.	Coverage	Selective acc.	Decision utility	Brier ↓
Trusted-latest	0.853	0.631	0.908	0.478	0.129
Time-aware-latest	0.906	0.611	1.000	0.572	0.080
PBS	0.932	0.824	0.977	0.768	0.060

The strong time-aware baseline is perfectly accurate among the subset on which it acts at threshold 0.70, but it acts on only 61.1% of queries. PBS covers 82.4% while retaining 97.7% selective accuracy.

Threshold-free AURC is **0.0107 for PBS**, compared with **0.0208** for time-aware-latest and **0.0538** for trusted-latest.

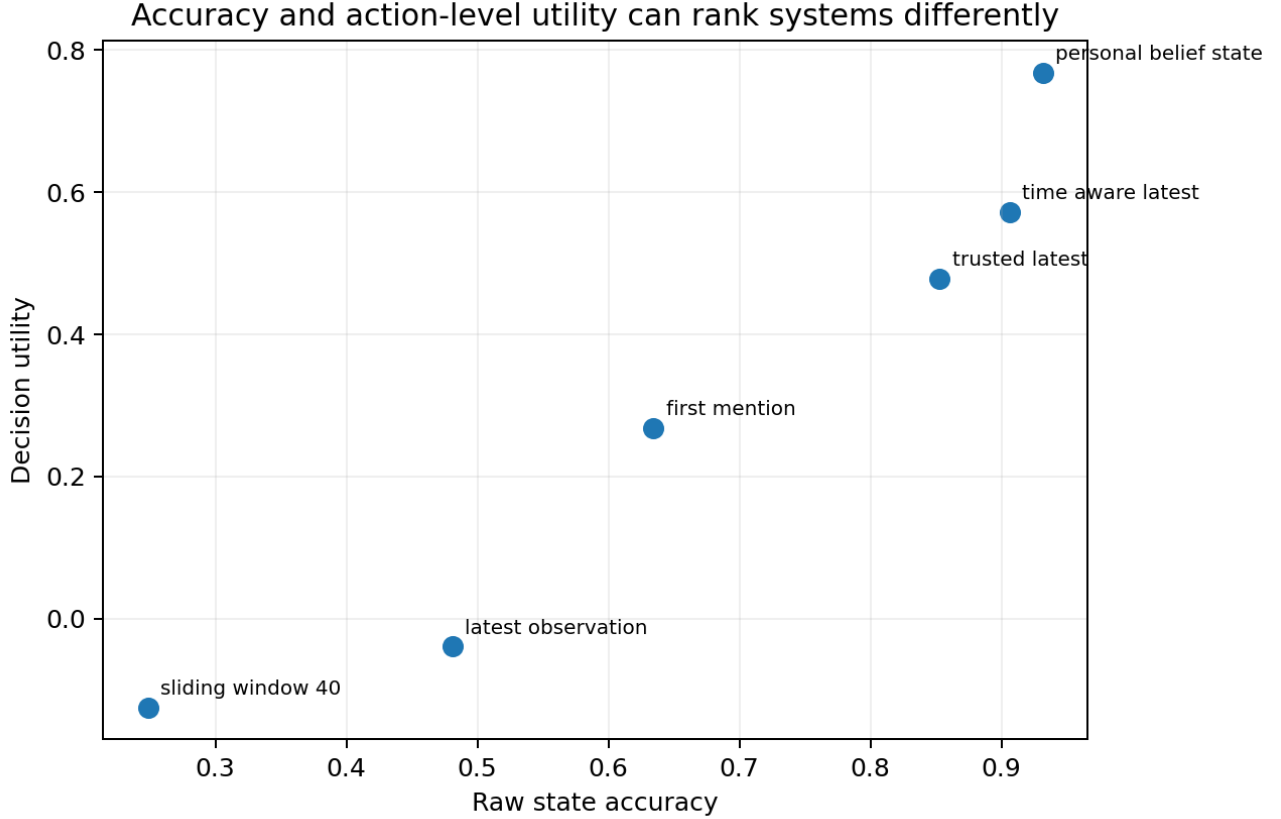


Figure 1: Raw state accuracy and decision utility on the standard controlled benchmark.

7.2 Held-out standard seeds

PBS hyperparameters are frozen from development seeds 1–3. Across test seeds 10–19, relative to time-aware-latest:

PBS — time-aware-latest	Mean	95% bootstrap interval
Decision utility	+0.210	[+0.205 , +0.216]
State accuracy	+0.024	[+0.021 , +0.026]
Selective accuracy	−0.020	[−0.022, −0.018]
Brier reduction	+0.020	[+0.019 , +0.021]

The selective-accuracy decrease is expected: PBS deliberately covers many more cases rather than only high-confidence explicit observations.

7.3 Long-horizon native-confidence diagnostic

With each tracker using its **native, unmatched confidence scale**, 800-turn histories produce an apparent ranking disagreement. In the checked-in shift run, time-aware-latest has slightly **higher** raw state accuracy than PBS (0.844 vs. 0.831) but lower fixed-threshold decision utility (0.207 vs. 0.503). Across ten held-out long-horizon seeds:

PBS — time-aware-latest	Mean	95% bootstrap interval
State accuracy	−0.014	[−0.017, −0.012]
Native-confidence decision utility	+0.284	[+0.276, +0.291]
Coverage	+0.328	[+0.322, +0.333]
Brier reduction	+0.016	[+0.014, +0.017]

This comparison is intentionally treated as a **diagnostic, not a fair model-selection result**, because the systems’ confidence scales have not yet been equalized. Section 7.6 applies the same development-only calibration opportunity to every tracker and shows that the fixed-threshold utility ordering reverses. The stable conclusion is therefore about protocol sensitivity, not a universal PBS utility advantage.

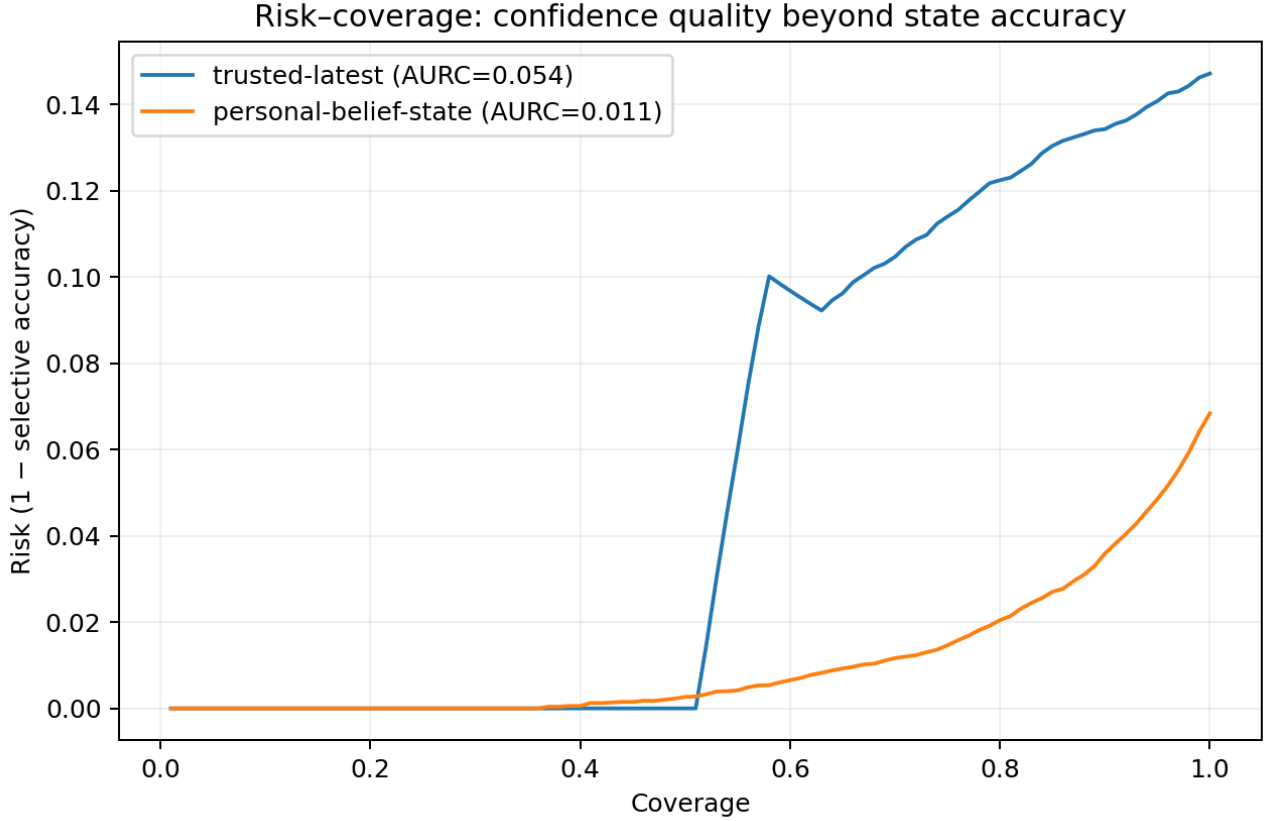


Figure 2: Risk–coverage curve for the strongest controlled systems.

7.4 Distribution shifts

PBS has the highest checked-in decision utility in all five regimes. Its utility remains 0.736 under high noise, 0.705 under implicit-heavy histories, 0.773 under temporary-heavy histories, and 0.503 at 800 turns.

7.5 Additional robustness analyses

Sensitivity sweeps, component ablations, a 48-cell stress grid, and exact paired tests are reported in the technical appendix and released artifacts. Briefly, PBS is preferred to time-aware-latest in **23/30** threshold/cost cells before shared calibration; provenance reliability and evidence aggregation are the largest PBS ablation effects; and PBS has higher utility in **46/48** controlled stress cells but is not uniformly superior. These analyses narrow rather than broaden the claim: operating-point conclusions depend on calibration and cost assumptions, while mechanism

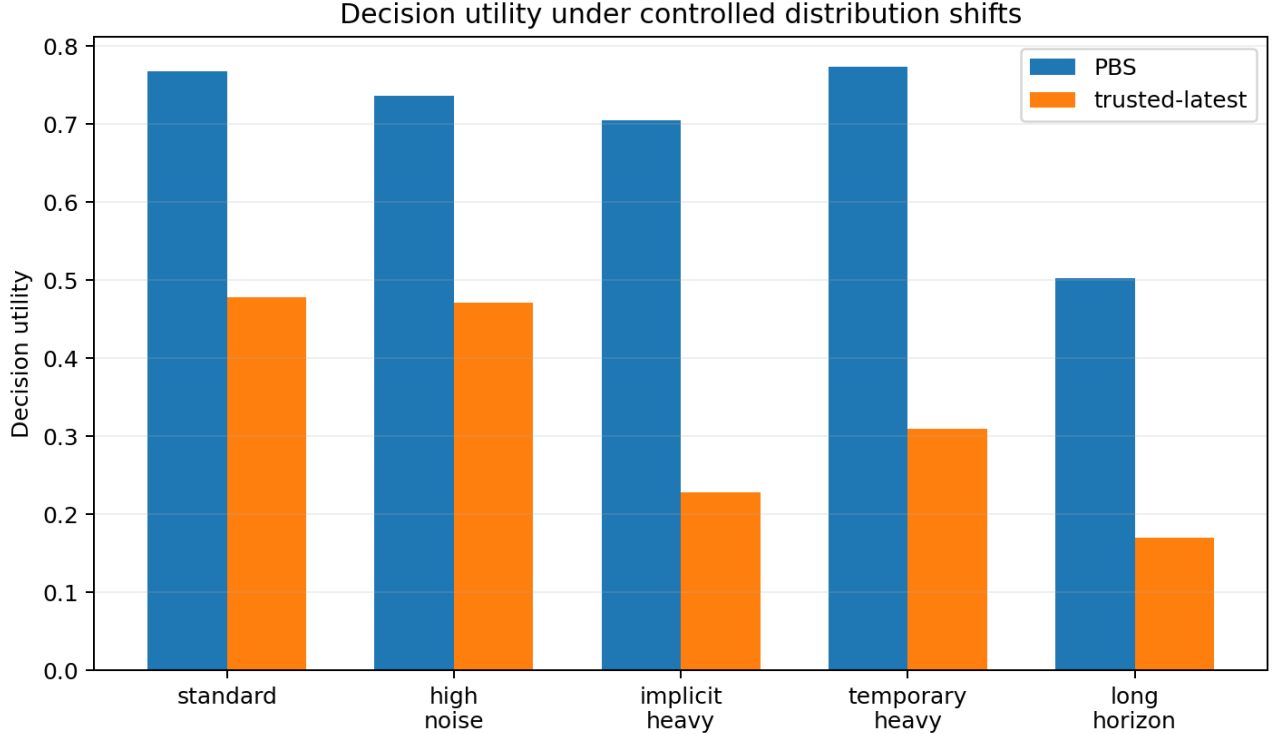


Figure 3: Decision utility under controlled distribution shifts.

effects are generator-dependent. A supplementary trajectory-level agent-evaluation extension also tests whether graders distinguish successful final states from policy-compliant, verified trajectories; it is reported in Appendix G and is not used as a frontier-model capability claim. Appendix I further evaluates the frozen grader on five held-out attack families, and Appendix J adds paired counterfactual checks of belief-state robustness versus adaptivity.

7.6 Equal-opportunity confidence calibration changes the conclusion

A fixed action threshold is only comparable when confidence scales are comparable. We therefore fit the same deliberately low-capacity post-hoc calibrator for **every tracker** on development seeds 1–3 only: each raw-confidence bucket is mapped to Laplace-smoothed empirical correctness, then frozen for held-out evaluation. No test labels are used.

On 800-turn held-out seeds, this changes the conclusion. The calibrated time-aware-latest baseline obtains **0.687 decision utility**, compared with **0.588 for PBS**. PBS still has higher selective accuracy (**0.908 vs. 0.844**) and slightly lower Brier (**0.119 vs. 0.122**), but its lower coverage (**0.750 vs. 1.000**) is penalized at the chosen deferral cost. Thus the earlier native-confidence utility advantage (**+0.284** on average) is not protocol-invariant; after shared calibration the sign reverses to approximately **−0.099**.

This is a central negative result. It does **not** imply that time-aware-latest is globally better, nor that PBS is globally better. It implies that fixed-threshold utility combines three choices—confidence scale, action threshold, and deferral cost—and can therefore change system ordering when those choices change. AURC is less exposed to this particular confound because monotone post-hoc calibration does not alter confidence ordering; in the checked-in benchmark AURC remains **0.0107 for PBS vs. 0.0208 for time-aware-latest**.

We therefore recommend a reporting hierarchy: (1) state accuracy, (2) calibration error, (3) risk-coverage/AURC, and (4) operating-point utility only after an explicitly shared calibration protocol. The complete mappings and held-out results are released in `reports/CALIBRATED_BASELINES.md` and `reports/CALIBRATION_PROTOCOL_ANALYSIS.md`.

A broader frozen protocol sweep reaches the same conclusion without relying on a single threshold or calibrator.

The focused PBS-vs-time-aware slice reverses frequently across 75 cells, and the full audit ranks **all six released baselines** under the same protocol family. Time-aware-latest wins 45 cells, PBS 27, and first-mention 3. Across unordered pairs of protocol cells, the probability of selecting a different winner is **0.516**; mean complete-ranking Kendall τ is **0.699**, with a minimum of **0.20**. To test whether this headline is a seed or grid artifact, a separate 10-user-per-seed diagnostic runs **2,000 paired seed bootstraps**, yielding winner-change probability **0.516** [**0.507, 0.538**] and mean Kendall τ **0.702** [**0.692, 0.720**]. Thirteen leave-one-level-out grid perturbations all retain multiple winners; winner-change probability ranges **0.327–0.564**. Appendix K reports the focused matrix, Appendix L the full-leaderboard and benchmark-health audit, and Appendix M the seed/grid robustness checks. These intervals quantify synthetic-generator seed variation only; they are not population estimates or external leaderboard results.

8. Relation to prior work

The personalization benchmark landscape is now crowded, so PersonaMetrica does **not** claim to introduce the first benchmark for preference following, evolving preferences, personalized memory, or proactive assistance. PrefEval evaluates explicit and implicit preference following in long-context conversations (S. Zhao et al. 2025). PersonaLens evaluates personalized task-oriented assistants with simulated users and LLM judges (Z. Zhao et al. 2025). HorizonBench directly studies evolving preferences over six-month synthetic histories and diagnoses stale-belief failures (Li et al. 2026). BeliefShift evaluates temporal belief consistency, contradiction resolution, and evidence-driven revision (Myakala, Agrawal, and Manche 2026). PERMA models event-driven preference evolution in realistic task environments (Liu et al. 2026). PerMemBench studies personalized memory-storage policies (In et al. 2026), while PAHF studies continual personalization with pre-action clarification and post-action feedback (Liang et al. 2026). pi-Bench evaluates proactive personal assistants with hidden intents, tools, and persistent sessions (Zhang et al. 2026). OP-Bench targets over-personalization such as irrelevance, repetition, and sycophancy (Hu et al. 2026).

PersonaMetrica targets a different layer: **measurement validity when personalization systems emit confidence and that confidence gates action**. Its core empirical object is not a new memory leaderboard but *ranking stability under defensible evaluation changes*. Native-confidence utility and shared-calibration utility can select different systems at the same nominal action threshold, while threshold-free risk-coverage tells a different story again. The trajectory extension then applies the same skepticism to the evaluator: a grader that is perfect on authored perturbations can fail catastrophically on held-out attack families, motivating causal-contract checks, **Unknown** under missing evidence, grader fingerprints, and disagreement review.

This positioning is deliberately narrower than a “first” claim. Related work in selective prediction already establishes risk-coverage and abstention as general evaluation tools, while 2026 work studies ranking instability under prompt/evaluator variation (Du et al. 2026; Mahmood, Abdaljalil, and Kurban 2026), disagreement-aware stable aggregation (Bonagiri et al. 2026), long-horizon trajectory attribution (Chen et al. 2026), and trajectory-level safety calibration under adaptive attacks (Asif et al. 2026). PersonaMetrica therefore does **not** claim that rank instability, trajectory attacks, or grader auditing are individually new. Its narrower contribution is to connect protocol-sensitive system selection with **evolving personal state**, confidence-gated action, counterfactual preference interventions, and evaluator validity in one reproducible artifact.

9. Human and model evaluation protocol

The release contains **620 independently labelable tasks** (300 proactivity decisions, 320 belief-update decisions) with a browser interface that hides simulator reference labels, plus a provider-neutral external-model evaluator reporting accuracy, Brier, and ECE. It also includes an executable adapter for the official HorizonBench per-item result JSONL contract; the checked-in adapter run uses synthetic schema fixtures only. These are **unexecuted external-evidence infrastructure**: no human or frontier-model result is reported. The intended next study holds the evaluation protocol fixed while replacing simulator-derived evidence or labels with independent human/model outputs.

10. Limitations

PersonaMetrica-Bench begins **after structured evidence extraction**, uses synthetic users, binary preference domains, designed source reliabilities, and generator-defined transition processes. PBS parameters are selected on

development simulator seeds and should not be interpreted as learned human preference dynamics. The benchmark can exhibit generator–method coupling, and multilingual tags are metadata rather than multilingual understanding results. No frontier-model table or human-subject study is reported. Decision utility is task-dependent, and this paper directly shows that its ranking can reverse after shared confidence calibration; utility must therefore be interpreted only with calibration, threshold, and deferral cost specified. Finally, the current Croissant metadata and anonymous bundle are complete locally, but venue-compliant external anonymous hosting remains a submission requirement.

11. Reproducibility and responsible release

All controlled primary experiments run on CPU; the release records seeds, generated data, scripts, figures, compute measurements, integrity audits, a dataset card, Croissant metadata, and an anonymous submission bundle. The benchmark contains only synthetic users. Because personal-state systems can enable persistent profiling or overconfident action, the broader system artifact includes forgetting, privacy-policy hooks, permission boundaries, provenance, and outcome verification. No human-subject result supports the present claims.

The practical evaluation implication is simple: when personal state can trigger action, report **state accuracy, calibration, risk–coverage/AURC, and any operating-point utility together**, and apply a shared confidence calibration protocol before comparing fixed-threshold action policies.

12. Conclusion

Personalization is a belief-update problem before it is a retrieval problem, and an action problem after it. PersonaMetrica-Bench shows that system rankings can change not only when moving from state accuracy to action metrics, but also when the **confidence calibration and action protocol itself** changes. In the controlled long-horizon setting, native-confidence utility favors PBS, whereas equal development-only post-hoc calibration can favor time-aware-latest at the same operating point; threshold-free AURC still favors PBS. Across the full six-system 75-cell audit, even the leaderboard winner and complete ordering vary substantially. The larger contribution is therefore evaluative rather than algorithmic: long-horizon personal assistants should be compared with explicit calibration protocols, threshold-free risk metrics, and ranking-stability diagnostics rather than a single operating-point leaderboard. PBS is one transparent reference implementation for studying those interactions, not a claim of a universally superior memory architecture.

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Technical appendix

Appendix A. Benchmark integrity audit

The standard generator was independently re-audited from source in v4.1.0. It deterministically produces **27,133 observations and 13,500 queries** for 500 users at seed 31. Replaying the first 25 users produces an identical SHA-256 digest. Exact structured fingerprints repeat across users because the benchmark intentionally uses a finite schema; the scientific unit is the evolving user history, not textual novelty. Full source/query distributions are in `reports/BENCHMARK_INTEGRITY_AUDIT.md`.

Appendix B. Experimental details

Development seeds are 1–3; standard held-out test seeds are 10–19; long-horizon held-out seeds are 30–39. PBS half-life and temperature are selected only by development Brier score. The action threshold used in the main operating-point table is 0.70 and deferral cost is 0.10; the sensitivity grid reports thresholds 0.50–0.90 and deferral costs 0–0.60. All component ablations reuse frozen PBS hyperparameters.

Appendix C. Exact metric definitions

For query i , let $y_i \in \{0, 1\}$ indicate whether the predicted current state is correct and let $c_i \in [0, 1]$ be the system confidence assigned to that prediction. For an action threshold τ , the system acts when $c_i \geq \tau$ and otherwise defers. We report:

- **State accuracy:** $N^{-1} \sum_i y_i$.
- **Coverage:** $N^{-1} \sum_i \mathbf{1}[c_i \geq \tau]$.
- **Selective accuracy:** $\sum_i y_i \mathbf{1}[c_i \geq \tau] / \sum_i \mathbf{1}[c_i \geq \tau]$, defined as zero when coverage is zero.
- **Brier score:** $N^{-1} \sum_i (c_i - y_i)^2$.
- **Decision utility:** correct action +1, incorrect action −1, and deferral $-d$, averaged over all queries, where the main operating point uses $d = 0.10$.
- **Risk–coverage curve:** for each target coverage, queries are sorted by decreasing confidence and risk is 1–accuracy on the retained prefix. **AURC** is the numerical area under this curve; lower is better.

The paper treats state accuracy, Brier, and AURC as protocol-level metrics. Fixed-threshold utility is secondary because it additionally depends on confidence calibration, τ , and d .

Appendix D. Equal-opportunity confidence calibration

To test whether native confidence gives some trackers an unfair advantage, every tracker receives the same development-only post-hoc calibration opportunity. Raw confidence values are rounded to two decimals; for each value, development correctness is converted to a Laplace-smoothed empirical probability $(k + 1)/(n + 2)$. The mapping is fit on development seeds 1–3 only and frozen before standard held-out seeds 10–19 and long-horizon

seeds 30–39. Unseen raw-confidence values map to the nearest development bucket. This calibrator is intentionally low capacity: it removes the easiest confidence-scale mismatch without introducing a learned model.

The long-horizon utility ordering changes after this calibration: native confidence gives PBS a mean +0.284 utility advantage over time-aware-latest, while shared development calibration yields approximately -0.099. The paper therefore does not use the native-confidence utility ranking as evidence of universal method superiority.

Appendix E. Claim-to-evidence matrix

Claim	Primary evidence	Important exclusion
State accuracy alone is incomplete for acting assistants	standard + 800-turn comparisons; risk-coverage	synthetic structured evidence only
Fixed-threshold utility is calibration-sensitive	development-calibrated baseline analysis	simple bucket calibrator; no frontier models
PBS confidence ordering is stronger in the checked-in benchmark	AURC 0.0107 vs. 0.0208	does not imply better utility at every operating point
PBS behavior depends strongly on provenance/aggregation	held-out component ablations	effect sizes are generator-dependent
Findings persist across many controlled regimes	48-cell stress grid	two cells favor the baseline

Appendix F. Additional robustness analyses moved from the main text

7.5 Sensitivity to action threshold and deferral cost

At the tested 180-user sensitivity seed, PBS has higher utility than trusted-latest in **30/30** threshold/cost cells and higher utility than time-aware-latest in **23/30** cells. The remaining cells matter: utility is not a property of the memory system alone but of the decision rule and the cost assigned to deferral. For this reason, the paper treats risk-coverage/AURC as primary and single-threshold utility as an operating-point analysis.

7.6 Which PBS components matter?

We ablate PBS components on held-out seeds 10–19 (160 users $\times$ 240 turns per seed), without retuning the remaining components. Removing recency lowers decision utility by **20.7 pp**; replacing provenance-aware source reliability with uniform reliability lowers it by **71.0 pp**; and removing evidence aggregation lowers it by **74.6 pp**. Temporary expiry and context scope have smaller but measurable effects on this generator.

Variant	State acc.	Coverage	Decision utility	Brier $\downarrow$
Full PBS	0.937	0.831	0.779	0.058
No recency	0.787	0.858	0.572	0.149
No provenance reliability	0.537	0.830	0.070	0.376
No temporary expiry	0.932	0.821	0.765	0.061
No context scope	0.928	0.829	0.761	0.065
No evidence aggregation	0.479	0.121	0.033	0.150

The large provenance and aggregation effects are also a warning about benchmark-method coupling: PersonaMetrica-Bench explicitly varies evidence source and contradiction structure, so methods that ignore those fields are expected to degrade. The benchmark therefore treats these ablations as mechanism diagnostics, not evidence that the same effect sizes will hold on natural conversations.

7.7 Stress grid and failure region

We additionally sweep **48 controlled cells**: four distractor-noise rates $\times$ four behavioral-evidence error rates $\times$ three temporary-preference rates, with 90 users $\times$ 240 turns per cell. PBS has higher decision utility than

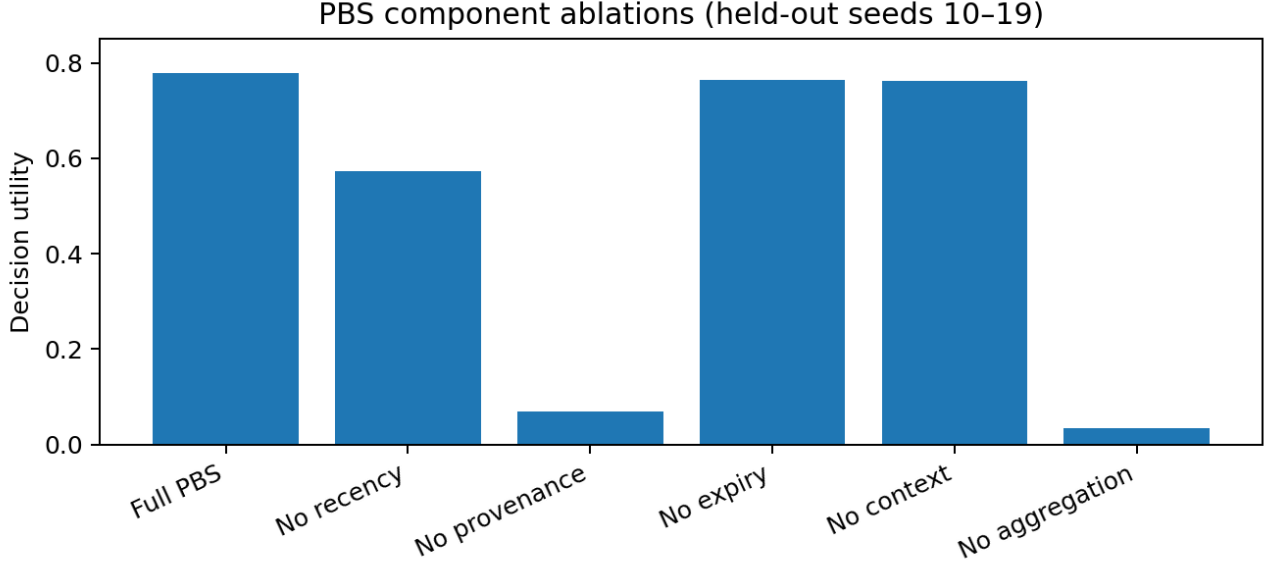


Figure 4: PBS component ablations on held-out simulator seeds.

time-aware-latest in **46/48** cells and higher raw state accuracy in **34/48**. Mean utility gain is **+0.138**, but the worst cell is **−0.025**. This failure region matters: PBS is not uniformly superior when source reliabilities are strongly misspecified.

This result narrows the claim: PBS is a useful reference implementation for the proposed evaluation framing, not a universally optimal state estimator.

7.8 Paired held-out tests

The primary ten-seed comparisons are paired by simulator seed. Exact two-sided sign tests and exact sign-flip permutation tests give $p = 0.00195$ for the standard utility gain, standard accuracy gain, long-horizon utility gain, long-horizon accuracy reversal, and long-horizon Brier reduction. All ten long-horizon seeds show higher PBS utility while all ten show lower PBS raw state accuracy. These tests quantify consistency across the pre-specified simulator seeds; they are **not** inferential claims about a human population. More importantly, these native-confidence utility tests do not by themselves establish a fair action comparison across systems; Section 7.6 repeats the comparison after a shared development-only calibration protocol.

The corresponding machine-readable outputs are `data/belief_sensitivity.json`, `data/pbs_component_ablations.json`, `data/belief_stress_grid.json`, and `data/belief_paired_tests.json`.

Appendix G. Trajectory-aware agent evaluation extension

The release includes a controlled agent-evaluation extension using the task/trial/grader/trajectory decomposition. **Eighty task specifications × ten trials = 800 trajectories** span reminders, notes, and task creation. Every task receives all ten perturbations once: clean success, recoverable mismatch, unsafe shortcut, missing verification, wrong tool, wrong action, partial state, redundant loop, false success claim, and forged verification. The suite is diagnostic rather than a frontier-model benchmark.

An end-state-only grader achieves **68.3% accuracy** and falsely accepts **44.3%** of gold-failing trials. A trajectory-policy grader that checks permission, tool/action constraints, bounded execution, and a self-reported verification-success flag reaches **90.0% accuracy** but still has **13.9% false accepts**. The robust trajectory grader recomputes verification from recorded expected/observed state and reaches **0% false accepts** on the authored suite.

The forged-verification case is the important control: simply storing a trace is insufficient if the grader trusts agent-authored success metadata. The evaluation needs independently checkable state evidence. Full results are in

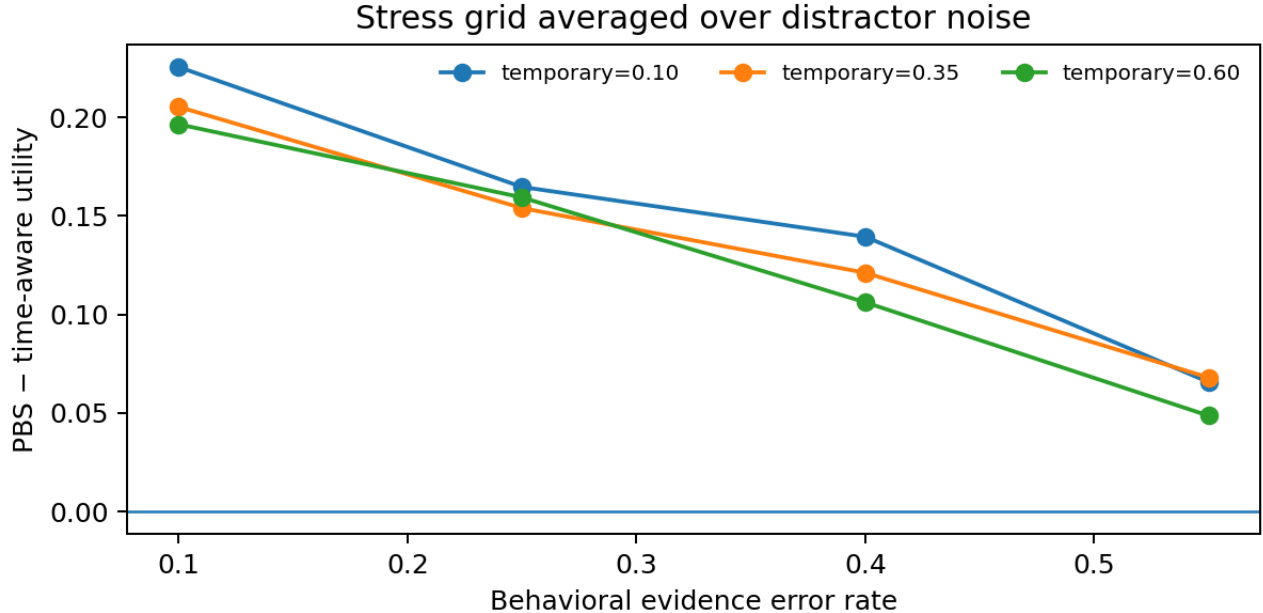


Figure 5: Stress grid averaged over distractor-noise levels.

reports/AGENT_TRAJECTORY_EVAL.md and data/agent_trajectory_eval.json.

The same run exports **800 trajectory records** and **254 failure-derived correction pairs**. Those text pairs are useful as data-flywheel plumbing but are highly templated; reports/TRACE_FAILURE_DATA_AUDIT.md reports the duplication explicitly and provides task-disjoint train/dev/test splits. Structured corrections with the trace evidence and failed assertions are separately exported in data/trajectory_corrections.jsonl. No post-training generalization claim is made from these artifacts.

Appendix H. Grader reliability, drift, and review routing

The evaluator itself is stress-tested rather than assumed correct. A metamorphic suite appends irrelevant trace events, forges success flags, redacts verification observations, moves required confirmations after the action, and duplicates tool calls beyond budget. The robust grader is **100% invariant** to benign events and forged success bits, detects all tested late-confirmation and excessive-loop mutations on applicable tasks, and an evidence-aware variant **abstains on 100%** of traces whose critical verification observation is redacted.

A weighted composite grader also exhibits threshold drift: on the checked-in controlled suite, lowering the pass threshold from 0.95 to 0.55 raises false accepts from roughly **14% to 58%**. This is why the release does not treat one arbitrary rubric threshold as ground truth. Repeated-trial analysis also shows that task-level estimate variance shrinks with more trials per task (sampling SD about **4.1 pp with one trial vs. 1.5 pp with eight trials** in the checked-in reliability run).

All grader disagreements are exported to data/grader_disagreement_queue.jsonl, with a blinded human-calibration UI in annotation/trajectory_grading/. The exact grader/task implementation is content-fingerprinted in data/grader_fingerprint.json; evaluator-code changes require regenerated results. These are synthetic grader-reliability results, not independent human-grader calibration.

Appendix I. Held-out evaluator attacks

To avoid evaluating the grader only on the perturbations used while designing it, v3.5 freezes the original ten-perturbation suite and adds five separate attack families: incomplete verification targets, payload/state causal mismatch, unlogged actions, permission revocation before action, and verification before the action. On 75 held-out authored attacks, the end-state grader falsely accepts **100%**, the v3.4 robust trajectory grader falsely accepts

96%, and the stricter causal-contract grader falsely accepts **0%** while retaining **100% accuracy on the original 800-trajectory gold suite**. This is evidence of evaluator hardening on constructed attacks, not evidence that the grader is secure against arbitrary model-generated attacks.

Appendix J. Counterfactual belief-state checks

We additionally run 540 paired structured-evidence interventions. Time-aware-latest and PBS are both highly sensitive to an explicit current update (100% and **99.3%**, respectively). Time-aware-latest flips on weak conflicting behavioral evidence in **90.0%** of pairs, whereas PBS flips in **35.6%**. PBS is invariant to a high-confidence other-person distractor in **98.0%** of pairs, leaving a small but concrete **2.0% nuisance-sensitivity failure rate**. These diagnostics expose a robustness/adaptivity tradeoff and are not natural-language understanding results.

Appendix K. Protocol Stability Matrix

To test whether the central rank-reversal result depends on a single hand-picked operating point, we freeze a broader protocol grid before interpreting the long-horizon diagnostic cohort: three confidence treatments (native, development-only bucket calibration, development-only isotonic calibration), five action thresholds, and five deferral costs, for **75 protocol cells**. Calibrators are fit on development seeds 1–3 only. The diagnostic cohort uses 40 users for each of long-horizon seeds 30–39; the larger primary benchmark remains unchanged.

PBS has higher mean utility in **30/75** cells and time-aware-latest in **45/75**. The probability that two distinct protocol cells select opposite pairwise winners is **0.486**; winner entropy is **0.971 bits**. Native confidence favors PBS in 20/25 cells, whereas each development-calibrated family favors time-aware-latest in 20/25 cells. This is a measurement-fragility result, not a claim that one calibration family is inherently correct or that the same reversal frequency holds for frontier models.

Protocol stability: PBS minus time-aware utility

		thr 0.55	thr 0.65	thr 0.70	thr 0.75	thr 0.85
native	cost 0.00	-0.040	+0.304	+0.256	+0.201	+0.091
	cost 0.05	-0.047	+0.324	+0.272	+0.213	+0.096
	cost 0.10	-0.053	+0.344	+0.289	+0.226	+0.102
	cost 0.20	-0.066	+0.385	+0.322	+0.250	+0.112
	cost 0.35	-0.085	+0.447	+0.371	+0.288	+0.128
bucket-laplace	cost 0.00	-0.042	-0.046	-0.065	-0.093	+0.298
	cost 0.05	-0.045	-0.053	-0.076	-0.107	+0.318
	cost 0.10	-0.049	-0.060	-0.086	-0.121	+0.338
	cost 0.20	-0.056	-0.075	-0.108	-0.150	+0.378
	cost 0.35	-0.066	-0.096	-0.139	-0.192	+0.438
isotonic-laplace	cost 0.00	-0.046	-0.046	-0.084	-0.093	+0.289
	cost 0.05	-0.054	-0.054	-0.098	-0.107	+0.308
	cost 0.10	-0.062	-0.062	-0.111	-0.121	+0.327
	cost 0.20	-0.078	-0.078	-0.138	-0.150	+0.365
	cost 0.35	-0.102	-0.102	-0.177	-0.192	+0.423

Blue: PBS higher utility. Tan: time-aware-latest higher. Values are mean delta across seeds 30-39.

Figure 6: Protocol Stability Matrix. Values are mean PBS minus time-aware-latest utility across long-horizon diagnostic seeds; sign changes indicate protocol-dependent winner changes.

The exact grid is stored in `configs/protocol_registry.json`, fingerprinted in `data/protocol_registry_fingerprint.json`, and validated by `scripts/validate_protocol_registry.py`. This is a **release-level protocol lock**, not an externally timestamped preregistration. Full results are in `reports/PROTOCOL_STABILITY_MATRIX.md` and `data/protocol_stability_matrix.json`.

Appendix L. Full-Leaderboard Stability and Benchmark Health

The focused pairwise protocol sweep can understate evaluation instability because a third method may become competitive under a different confidence/cost regime. We therefore replay **all six released baselines** on the

same long-horizon diagnostic histories across the frozen 75-cell protocol family. Calibration is fit on development seeds 1-3 and frozen before long-horizon seeds 30-39.

Full-leaderboard quantity	Value
Time-aware-latest winner cells	45 / 75
PBS winner cells	27 / 75
First-mention winner cells	3 / 75
Probability two protocol cells choose different winners	0.516
Mean Kendall τ across complete rankings	0.699
Minimum Kendall τ	0.20

The three first-mention wins are treated as a **warning**, not a positive result for that baseline: an operating protocol can reward coverage/confidence economics strongly enough that a stale estimator becomes the apparent winner. This is why PersonaMetrica reports threshold-free state/risk metrics alongside operating-point utility.

Benchmark health is also release-gated. Synthetic identities are seed-scoped; development and test display IDs have zero overlap; after identifiers are removed, development and test complete-history fingerprints have zero overlap; the 500-user standard cohort has zero duplicate complete histories; all ten preference domains are queried; temporary-scope queries are present; answer skew remains bounded; and the benchmark is not saturated. Structured event fragments can repeat because the simulator uses a finite schema, but the evolving full-history unit does not duplicate in the checked-in standard cohort.

Appendix M. Seed uncertainty, grid robustness, and external-result contract

The full-leaderboard result is additionally stress-tested along two axes that were not used to choose the reported winner counts. First, a lower-cost diagnostic cohort with 10 users for each long-horizon seed replays all six methods across the same 75 protocol cells. We resample the ten long-horizon seeds **2,000 times as paired units**, using identical resampled seed indices for every method and protocol. Winner-change probability is **0.516** with a 95% seed-bootstrap interval of **[0.507, 0.538]**; mean complete-ranking Kendall τ is **0.702 [0.692, 0.720]**. In all bootstrap replicates, winner-change probability remains above 0.40. The least stable individual protocol-cell winner has bootstrap support **0.619**, so the release does not treat every cell winner as certain. This diagnostic uses fewer users per seed than the primary 40-user-per-seed leaderboard audit and is reported as robustness evidence, not a replacement primary estimate.

Second, we run **13 leave-one-level-out perturbations** of the frozen protocol grid: each calibration family, action threshold, and deferral cost is omitted once without inventing replacement levels from test outcomes. Every perturbed grid retains more than one winner. Winner-change probability ranges **0.327–0.564** and mean Kendall τ ranges **0.653–0.895**. Removing native confidence produces the largest stabilization, which reinforces rather than weakens the calibration-fairness interpretation: comparison on unmatched native confidence scales is a major source of apparent system selection.

Finally, the release includes `external_benchmarks/horizonbench_adapter.py`, which consumes the per-item JSONL result schema documented by the official HorizonBench evaluation workflow. Standard HorizonBench output can be summarized by overall/evolved/static accuracy; confidence-gated PersonaMetrica utility is enabled only when an explicit `confidence` field is supplied. The checked-in smoke test uses **synthetic schema fixtures only**. No HorizonBench score, downloaded benchmark example, or frontier-model result is claimed.

Full machine-readable outputs are in `data/leaderboard_seed_uncertainty.json`, `data/protocol_grid_robustness.json`, and `data/horizonbench_adapter_smoke.json`.

NeurIPS 2026 paper checklist

1. **Claims** — **Yes**. The Abstract, Introduction, Results, and Limitations explicitly scope claims to a controlled structured-evidence benchmark and separate executed results from future human/model experiments.
2. **Limitations** — **Yes**. Section 10 enumerates structured-evidence, synthetic-user, binary-domain, designed-reliability, generator-coupling, external-model, human-study, utility, and multilingual limitations.

3. **Theory, assumptions, and proofs** — **N/A**. The paper contains a probabilistic formulation but no theorem-level theoretical claim.
4. **Experimental result reproducibility** — **Yes**. Code, fixed seeds, full generated data, exact experiment scripts, statistical scripts, figures, and reproduction logs are included in the anonymized bundle.
5. **Open access to data and code** — **Not yet venue-compliant**. The anonymous supplementary bundle contains the code/data needed for the reported controlled results, but a persistent reviewer-accessible anonymous host has not been configured from this environment. This remains an explicit submission blocker and must be resolved before submission.
6. **Experimental setting/details** — **Yes**. Sections 3–7 and Appendices B–D specify splits, hyperparameter selection, baselines, metric definitions, thresholds, costs, confidence calibration, and held-out seeds.
7. **Statistical significance** — **Yes**. The main effects include bootstrap intervals over pre-specified simulator seeds and exact paired sign/sign-flip tests; the paper states what source of variation these quantify.
8. **Compute resources** — **Yes**. Section 11 and `reports/COMPUTE_RESOURCES.md` report CPU, RAM, wall time, and peak-memory examples.
9. **Code of Ethics** — **Yes**. The executed benchmark uses synthetic users only; no deceptive human data collection or restricted personal data are used.
10. **Broader impacts** — **Yes**. Section 11 discusses risks from persistent profiling, sensitive inference, and overconfident action, plus mitigations in the system artifact.
11. **Safeguards** — **N/A for model release**. No high-risk pretrained model is released. Privacy, forgetting, permission, provenance, and verification hooks are included in the broader system artifact.
12. **Licenses** — **Yes**. The project is released with a license and cites the research assets used for scientific comparison. No external dataset is repackaged into PersonaMetrica-Bench.
13. **New assets** — **Yes**. The benchmark, generator, dataset card, Croissant metadata, benchmark specification, code, and limitations are documented alongside the release.
14. **Crowdsourcing/human subjects** — **N/A for reported results**. No crowdsourced or human-subject data support the paper’s current empirical claims. The unexecuted annotation protocol is included for future work only.
15. **IRB/equivalent approval** — **N/A for reported results**. No human-subject experiment has been executed. Appropriate review would be obtained before running a study where required.